

Chemical Thermodynamics

Understanding Spontaneous Processes

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Outline

1. Spontaneity
2. Entropy
3. The Second and Third Laws of Thermodynamics
4. Free Energy

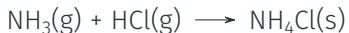
Spontaneity

Spontaneous and non-spontaneous processes

Definition

A **spontaneous process** occurs without outside intervention. A **nonspontaneous process** must be "driven" by an input of energy.

How can we predict whether a given process will be spontaneous?



Spontaneity and rate are not correlated

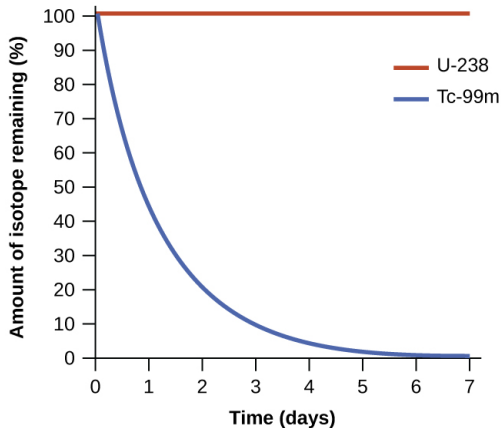


Figure 1: Spontaneous processes may be fast or slow.

Spontaneity derives from dispersal of matter and energy

Fundamental Criterion for Spontaneous Processes?

Spontaneous processes generally involve **dispersal of matter and/or energy**.

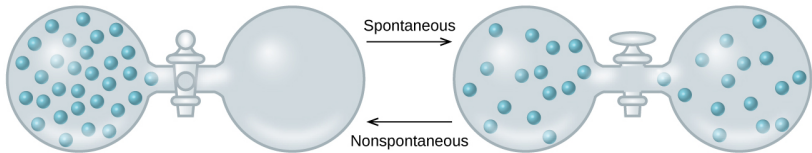


Figure 2: Expansion of a gas into vacuum in an isolated vessel is spontaneous.

Spontaneity derives from dispersal of matter and energy

Fundamental Criterion for Spontaneous Processes?

Spontaneous processes generally involve **dispersal of matter and/or energy**.



Figure 3: Heat flow from a hotter body to a colder body is spontaneous.

Spontaneity derives from dispersal of matter and energy

Redistribution of Matter during a Spontaneous Process

Describe how matter is redistributed when the following spontaneous processes takes place.

- a. A solid sublimates.
- b. A gas condenses.
- c. A drop of food coloring added to a glass of water forms a solution with uniform color.

Entropy

Reversible processes involve infinitesimal changes

Definition

In a **reversible** process, the system remains in equilibrium and changes only through infinitesimal steps (e.g., dx).

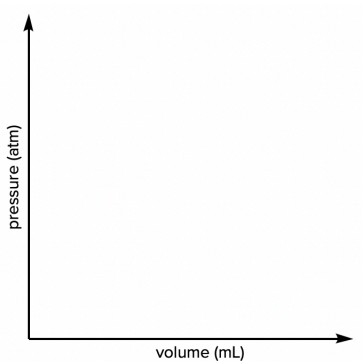


Figure 4: Reversible and irreversible isothermal expansion of an ideal gas.

Reversible processes involve infinitesimal changes

- Changes in state functions are the same for reversible and irreversible processes with the same initial and final states
- Changes in *path functions* differ; e.g. $q_{irrev} \neq q_{rev}$
- Reversible processes can be approximated but never achieved physically
- For much of thermodynamics, thinking about reversible processes is convenient

The classical definition of entropy

Definition

The change in **entropy** is defined as the heat for the process when conducted reversibly divided by the temperature at which it takes place: $\Delta S = \frac{q_{rev}}{T}$. For a process with changing T , $\Delta S = \int \frac{dq}{T}$.

- State function
- Measure of energy dispersal
- Increases when heat is transferred into a system

The classical definition of entropy

ΔS° for Vaporization of Water

Calculate ΔS° for the conversion of 1 mole of liquid water to water vapor at constant pressure. ΔH_{vap}° for water is 40.7 kJ/mol.

The probabilistic (statistical) definition of entropy

Microstates and Macrostates

The **microstate** of a system consists of all of the positions x_i and velocities v_i of particles within the system. The **macrostate** is defined by state functions of the bulk system: P, V, T, n, U, S , etc.

Statistical Entropy

How random are the set of microstates that correspond to a given macrostate?

The probabilistic (statistical) definition of entropy

- Macrostate S is a weighted average of microstate S 's:

$$S = \sum_i p_i f(p_i)$$

- For a composite system $A + B$ (proof):

$$S = \sum_i \sum_j p_i p_j f(p_i p_j) = \sum_i p_i f(p_i) + \sum_j p_j f(p_j)$$

- Thus, the function of p_i is the natural logarithm: $S = \sum_i p_i \ln p_i$
- Boltzmann's constant** k_B gives entropy units of energy per absolute temperature: $S = -k_B \sum_i p_i \ln p_i$

The probabilistic (statistical) definition of entropy

- Manipulate the argument of the logarithm: $S = k_B \ln \frac{1}{\prod_i p_i^{p_i}}$
- Define: $W = \frac{1}{\prod_i p_i^{p_i}}$, the number of microstates corresponding to the given macrostate (**likelihood**)
- We can now re-write the statistical definition of entropy using W :
 $S = k_B \ln W$

The probabilistic (statistical) definition of entropy

Determination of ΔS

Calculate the change in entropy for the process depicted below.



The sign of ΔS can be predicted

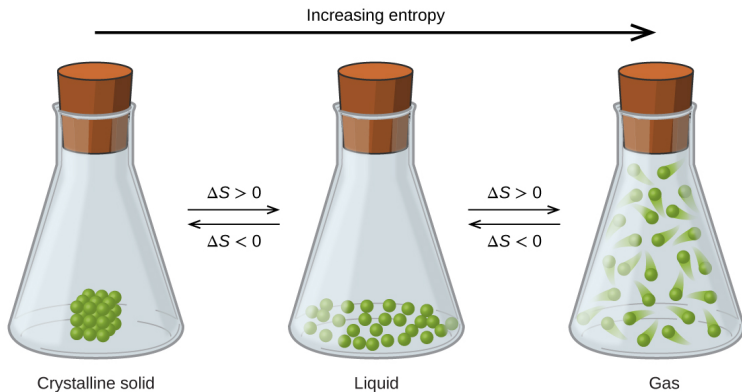


Figure 5: Entropy increases from solid to liquid to gas.

The sign of ΔS can be predicted

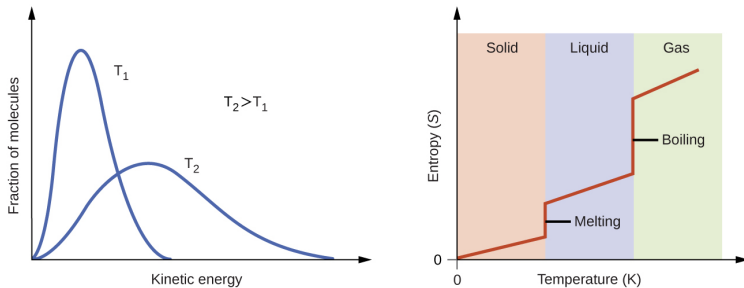


Figure 6: Entropy increases as absolute temperature increases.

The sign of ΔS can be predicted

Predicting the Sign of ΔS

Predict the sign of the entropy change for the following processes.

a. Heating 1 mol $\text{H}_2\text{O}(\text{l})$ from 20 °C to 50 °C

b. $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \longrightarrow \text{AgCl}(\text{s})$

c. $\text{C}_6\text{H}_6(\text{l}) + 7.5 \text{O}_2(\text{g}) \longrightarrow 6 \text{CO}_2(\text{g}) + 3 \text{H}_2\text{O}(\text{l})$

d. $\text{NH}_3(\text{s}) \longrightarrow \text{NH}_3(\text{l})$

The Second and Third Laws of Thermodynamics

The second law governs spontaneity

Entropy Change of the Universe

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr}$$

- Consider a system with $T_{sys} > T_{surr}$
- Heat q_{rev} flows *spontaneously* from hotter body to colder body:
$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = \frac{-q_{rev}}{T_{sys}} + \frac{q_{rev}}{T_{surr}} > 0$$
- If $T_{sys} = T_{surr}$, $\Delta S_{sys} + \Delta S_{surr} = 0$
- For *nonspontaneous* heat flow from colder body to hotter body,
 $\Delta S_{univ} < 0$
- The above observations generalize to any spontaneous process!

The second law governs spontaneity

The Second Law of Thermodynamics

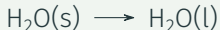
All spontaneous processes involve an increase (or no change) in the entropy of the universe: $\Delta S_{univ} \geq 0$.

Entropy Change	Spontaneity
$\Delta S_{univ} > 0$	spontaneous
$\Delta S_{univ} < 0$	nonspontaneous
$\Delta S_{univ} = 0$	at equilibrium

The second law governs spontaneity

Will Ice Spontaneously Melt?

The entropy change for the process below is 22.1 J/K and requires that the surroundings transfer 6.00 kJ of heat to the system. Is the process spontaneous at $-10.0\text{ }^{\circ}\text{C}$? What about $+10.0\text{ }^{\circ}\text{C}$?



The third law defines a zero of entropy

The Third Law of Thermodynamics

The entropy of a pure, perfectly crystalline substance at 0 K is zero.

$$S = k_B \ln 1 = 0.$$

Standard entropies enable calculation of ΔS°

Definition

The **standard molar entropy** (S°) of a substance is the absolute entropy of 1 mole of the substance under standard conditions (1.00 bar pressure and usually 298 K).

Calculating Standard Entropy Changes of Reaction

$$\Delta S^\circ = \sum_i^P n_i S_{f,i}^\circ - \sum_j^R n_j S_{f,j}^\circ$$

For example, for $m \text{ A} + n \text{ B} \longrightarrow x \text{ C} + y \text{ D}$,

$$\Delta S^\circ = [x S^\circ(\text{C}) + y S^\circ(\text{D})] - [m S^\circ(\text{A}) + n S^\circ(\text{B})]$$

Standard entropies enable calculation of ΔS°

Entropy depends on mass, phase, and state.

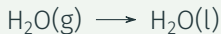
Substance	S° (J K ⁻¹ mol ⁻¹)
C(s, graphite)	5.74
C(s, diamond)	2.38
CO(g)	197.7
CO ₂ (g)	213.8
CH ₃ OH(l)	126.8
H ₂ (g)	130.57
H ₂ O(g)	188.71
H ₂ O(l)	69.91
O ₂ (g)	205.03

Table 1: Standard molar entropies for several substances.

Standard entropies enable calculation of ΔS°

Determination of ΔS°

Calculate the standard entropy change for the following process using tabulated standard entropies.



Standard entropies enable calculation of ΔS°

Determination of ΔS°

Calculate the standard entropy change for the combustion of methanol, CH_3OH .



Free Energy

A system-centric statement of the second law

- From the second law: $\Delta S_{univ} = \Delta S_{sys} + \frac{q_{surr}}{T} \geq 0$
- Heat transfer to the surroundings is the negative of the enthalpy change in the system (constant T and P): $\Delta S_{sys} - \frac{\Delta H_{sys}}{T} \geq 0$
- Rearrange the inequality: $\Delta H - T\Delta S \leq 0$

A system-centric statement of the second law

Gibbs Free Energy

Gibbs free energy (G) is defined as $H - TS$. It provides a concise statement of the second law. For any spontaneous process,

$$\Delta G \leq 0$$

Entropy Change	Spontaneity
$\Delta G < 0$	spontaneous
$\Delta G > 0$	nonspontaneous
$\Delta G = 0$	at equilibrium

At constant T and P , $\Delta G = \Delta H - T\Delta S$

What's "free" about ΔG ?

Change in Gibbs Free Energy

ΔG is the difference between the energy released by a process (ΔH) and energy lost to the surroundings ($T\Delta S$).

$\Delta G < 0$ Free energy change is the *maximum* energy available to do useful work

$\Delta G > 0$ Free energy change is the *minimum* energy required to drive the process

Calculating ΔG° : Standard enthalpies and entropies

Standard enthalpies of formation and standard entropies can be used as previously defined to calculate ΔG_{rxn}° .

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta H^\circ = \sum_i^P n_i \Delta H_{f,i}^\circ - \sum_j^R n_j \Delta H_{f,j}^\circ$$

$$\Delta S^\circ = \sum_i^P n_i S_{f,i}^\circ - \sum_j^R n_j S_{f,j}^\circ$$

Calculating ΔG° : Standard enthalpies and entropies

ΔG° from Standard Enthalpy and Entropy Changes

Use standard enthalpy and entropy data to calculate the standard free energy change for the vaporization of water at room temperature (298 K). Is the process spontaneous?

Calculating ΔG° : Standard free energies of formation

Standard Free Energy of Formation

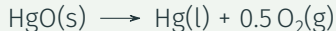
Standard enthalpies of formation and standard entropies of formation can be combined to calculate **standard free energies of formation** (ΔG_f°). These are often tabulated at 298 K.

$$\Delta G_f^\circ = \Delta H_f^\circ - T\Delta S_f^\circ$$

Calculating ΔG° : Standard free energies of formation

ΔG° from Standard Free Energies of Formation

Use standard free energies of formation to calculate ΔG° for the decomposition of mercury(II) oxide at 298 K. Is the reaction spontaneous?



Free energy changes follow Hess's law

Hess's Law

If a process can be written as the sum of several steps, the free energy change of the overall process is equal to the sum of the free energy changes of the steps.

$$R = R_1 + R_2 + \dots + R_N$$

$$\Delta G(R) = \sum_i^N \Delta G(R_i)$$

- Any reaction may be thought of as a set of **coupled reactions**
- Nonspontaneous reactions can be coupled to spontaneous reactions; e.g.,
 - $\text{ZnS(s)} \longrightarrow \text{Zn(s)} + \text{S(s)}, \Delta G^\circ = 201.3\text{kJ}$
 - $\text{S(s)} + \text{O}_2\text{(g)} \longrightarrow \text{SO}_2\text{(g)}, \Delta G^\circ = -300.1\text{kJ}$
 - $\text{ZnS(s)} + \text{O}_2\text{(g)} \longrightarrow \text{Zn(s)} + \text{SO}_2\text{(g)}$

Free energy changes follow Hess's law

Calculating Free Energy Change for a Coupled Reaction

Is a reaction coupling the decomposition of ZnS to the formation of H₂S spontaneous under standard conditions at 298 K? Use tabulated free energies of formation.

Spontaneity depends on temperature

$$\Delta G = \Delta H - T\Delta S$$

1. $\Delta H, \Delta S > 0$. Endothermic process with favorable increase in system entropy. Spontaneous above T_{crit}
2. $\Delta H, \Delta S < 0$. Exothermic process with unfavorable decrease in system entropy. Spontaneous below T_{crit}
3. $\Delta H > 0, \Delta S < 0$. Endothermic process with unfavorable decrease in system entropy. Nonspontaneous at all temperatures
4. $\Delta H < 0, \Delta S > 0$. Exothermic process with favorable decrease in system entropy. Spontaneous at all temperatures

Spontaneity depends on temperature

Summary of the Four Scenarios for Enthalpy and Entropy Changes

	$\Delta H > 0$ (endothermic)	$\Delta H < 0$ (exothermic)
$\Delta S > 0$ (increase in entropy)	$\Delta G < 0$ at high temperature $\Delta G > 0$ at low temperature Process is spontaneous at high temperature	$\Delta G < 0$ at any temperature Process is spontaneous at any temperature
$\Delta S < 0$ (decrease in entropy)	$\Delta G > 0$ at any temperature Process is nonspontaneous at any temperature	$\Delta G < 0$ at low temperature $\Delta G > 0$ at high temperature Process is spontaneous at low temperature

Figure 7: Spontaneity depends on temperature and the signs of ΔH and ΔS .

Spontaneity depends on temperature

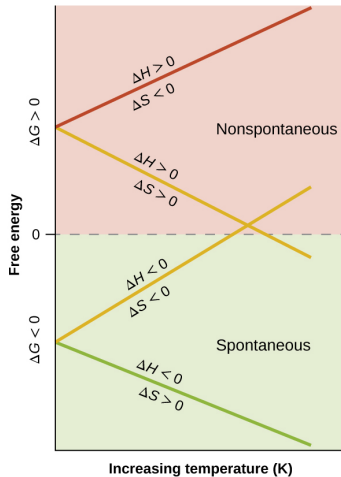
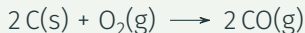


Figure 8: Dependence of free energy on temperature, ΔH , and ΔS .

Spontaneity depends on temperature

Predicting the Temperature Dependence of Spontaneity

The incomplete combustion of carbon is described by the following equation. How does the spontaneity of this process depend on temperature?



Calculating the temperature of a phase transition

Equilibrium Temperature for a Phase Transition

The boiling point of a liquid is the temperature at which the rates of vaporization and condensation are equal. Use tabulated enthalpies and entropies to estimate the boiling point of water under standard conditions.

ΔG° is for $Q_i = 1 \rightarrow Q_f = K$, standard state to equilibrium

Reaction Quotient

For a general reaction $a A + b B \rightarrow c C + d D$, $Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$.

The Standard Reaction Process

ΔG° refers specifically to the free energy change from an initial state of $Q = 1$ to a final state of $Q = K$.

	$r R$	$\rightleftharpoons$	$p P$
Initial	1.00		1.00
Equilibrium	$[R]_{eq}$		$[P]_{eq}$

Table 2: A general ICE table for the standard reaction process. The free energy change for this process is ΔG° .

From a non-standard state ($Q_i \neq 1$) to equilibrium

Non-standard Free Energy Change

Starting from a non-standard initial Q and ending at a final equilibrium state,

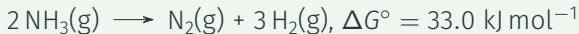
$$\Delta G = \Delta G^\circ + RT \ln Q$$

- As Q increases, the excess of products pushes ΔG more positive ("less spontaneous")
- As Q decreases, the excess of reactants pulls ΔG more negative ("more spontaneous")

From a non-standard state ($Q_i \neq 1$) to equilibrium

Calculating ΔG under Non-standard Conditions

What is the free energy change for the process below when $T = 298$ K, $P_{\text{N}_2} = 0.870$ atm, $P_{\text{H}_2} = 0.250$ atm, and $P_{\text{NH}_3} = 12.9$ atm?



ΔG° and K are profoundly related

The Law of Mass Action

For the Standard Reaction Process defined above,

$$\Delta G^\circ = -RT \ln K$$

- For a proof see: [Part 1](#), [Part 2](#)
- When $K > 1$, the standard free energy change is negative and the reaction is spontaneous starting from standard conditions
- When $K < 1$, the standard free energy change is positive and the reaction is nonspontaneous starting from standard conditions

ΔG° and K are profoundly related

Calculating an Equilibrium Constant Using ΔG°

Calculate the solubility product constant (K_{sp}) for AgCl using the standard free energies of formation of $\text{Ag}^+(\text{aq})$, $\text{Cl}^-(\text{aq})$, and $\text{AgCl}(\text{s})$.

Free energy is minimized in the equilibrium state

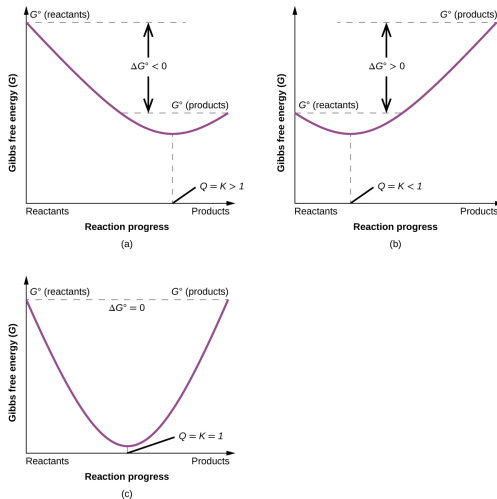


Figure 9: Free energy as a function of reaction extent for (a) $\Delta G^\circ < 0$, (b) $\Delta G^\circ > 0$, and (c) $\Delta G^\circ = 0$,